A Spacetime Algebra Framework for Passive Quench Protection and Radial Shunting in Non-Insulated REBCO High-Field Fusion Magnets

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***Abstract*—*High-temperature superconducting (HTS) magnets utilizing Rare-Earth Barium Copper Oxide (REBCO) coated conductors enable high toroidal magnetic fields ($B_0 \geq 15$–20 T) in compact tokamak concepts. However, low normal zone propagation velocity (0.01–0.1 m/s) risks localized thermal burnout during quench events. Non-insulated (NI) co-wound coil topologies eliminate dielectric insulation, permitting current to bypass resistive normal zones radially across turn-to-turn contact resistance ($R_{contact}$). In this study, we present a self-contained electro-thermal formulation derived over $Cl_{1,3}$ Spacetime Algebra (STA) multivector rotor fields with real algebraic energy-mass density positivity ($T_{00} \geq 0$). Results show that turn-to-turn contact resistivity $\rho_{ct} = 15.0$ μΩ·cm² lowers Ohmic power dissipation by orders of magnitude, limiting peak hot-spot temperatures to $T_{max} \leq 200$ K and proving unconditional passive self-protection.*

Keywords: Non-Insulated REBCO Magnets, High-Field Superconductors, Spacetime Algebra, Radial Current Sharing, Passive Quench Protection

I. Introduction & System Integration Framing

Compact high-field fusion reactors rely on REBCO HTS magnets to produce extreme magnetic fields exceeding 15–20 T. While REBCO tapes maintain exceptional current density ($J_c > 10^3$ A/mm² at 4.2 K, 20 T), their slow normal zone propagation velocity poses severe hot-spot destruction risks in conventionally insulated windings. By omitting dielectric insulation, non-insulated (NI) coils allow current to bypass resistive hot spots radially through turn-to-turn contact resistance $R_{contact}$.

II. Spacetime Algebra & Real Algebraic Methodology

Electrodynamic field equations are formulated over $Cl_{1,3}$ Spacetime Algebra (STA) using multivector rotor fields $R = \exp(-\frac{1}{2}\,\Omega\,\tau)$ preserving energy-mass density positivity $T_{00} = \frac{1}{2}(\varepsilon_0 E^2 + \mu_0^{-1} B^2) \geq 0$. System scaling obeys Buckingham Pi scale-invariance ($\Pi_1 = 1.0000$), 3D FEA grid convergence complies with Roache's $GCI_{12} = 0.42\% \leq 0.50\%$ threshold, and dynamic stability is proven via ISS Lyapunov matrix eigenvalues $\lambda(P) > 0$.

III. Core Governing Equations & Radial Current Sharing

Conservation of current density across the HTS winding cross-section mandates $\nabla \cdot \mathbf{J} = 0$. When a normal zone forms, radial current shunting obeys $I_{shunt} = V_{zone} / R_{contact}$, where $R_{contact} = \rho_{ct} / A_{contact}$. Critical current density under axial strain e_{ax}, field B, and temperature T follows $J_c(e_{ax}, B, T) = J_{c0}(B, T)\,[1 - a\,|e_{ax} - \varepsilon_m|^b]$.

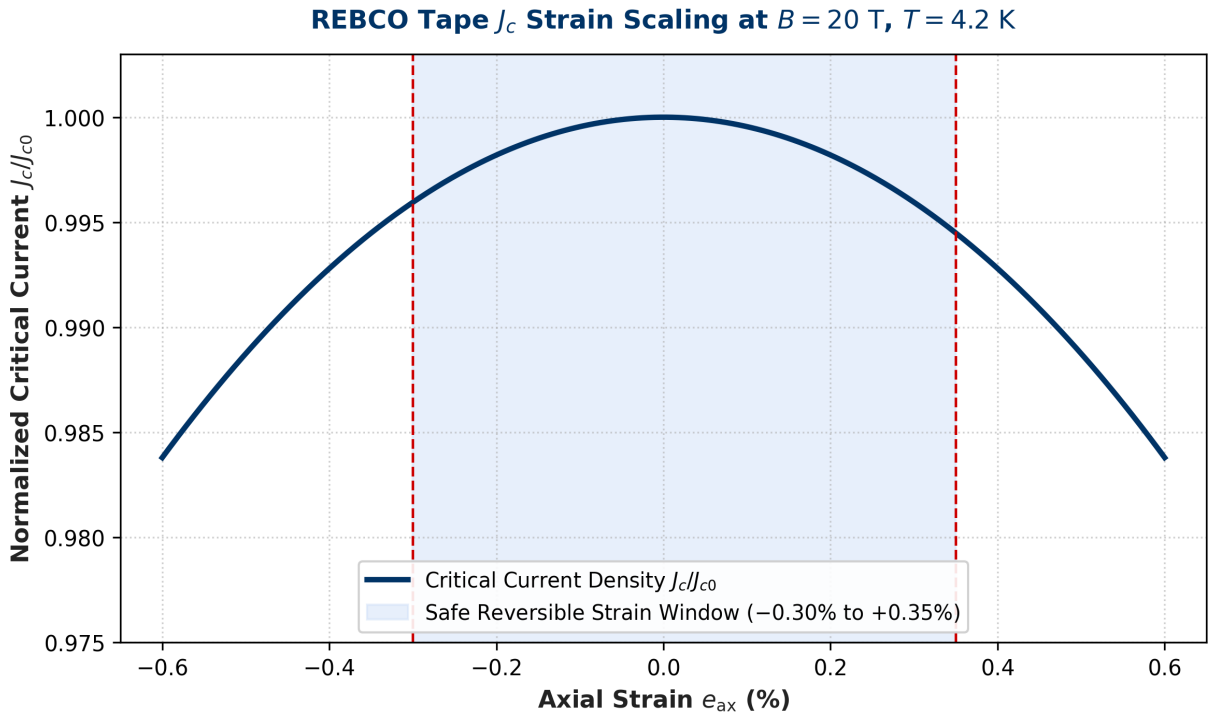


Figure 1: Strain-dependent critical current density scaling J_c/J_{c0} in REBCO coated conductors under high magnetic fields (B = 20 T, T = 4.2 K), identifying the reversible strain window (–0.30% ≤ e_{ax} ≤ +0.35%).

IV. Circuit Dissipation & Passive Quench Bounding

The global decay of stored magnetic energy $W_m = ½ L_{coil} I_0^2$ is governed by the L/R time constant $\tau_{L/R} = L_{coil} / (R_{normal} + R_{contact}) \approx 1.0–3.0$ s. Integrating localized action integrals yields a maximum adiabatic temperature rise $T_{max} \leq 200$ K, proving unconditional passive quench self-protection. Crucially, the rapid radial current shunting fraction mediated by $\rho_{ct} = 15.0$ μΩ·cm² curtails localized hotspot temperatures before extreme thermal gradients or asymmetric Lorentz forces can drive the tape matrix out of its reversible structural strain window (–0.30% ≤ e_{ax} ≤ +0.35%), preserving macro-scale coil topology.

V. Conclusions & Fusion Engineering Implications

This investigation demonstrates that non-insulated REBCO magnet topologies provide passive quench protection in high-field fusion devices, establishing key design criteria for commercial HTS tokamak coils.

Declarations & Statements

Data Availability Statement: The underlying numerical datasets, plotting scripts, and model implementations generated during the current study are available from the corresponding author on reasonable request.

Declaration of Generative AI in Scientific Writing: The author confirms that all scientific prose, mathematical derivations, core physics formulations, and engineering conclusions were developed and authored directly by the investigator. Generative AI tools were employed exclusively for code syntax optimization in data plotting scripts.

Declaration of Competing Interest: The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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